

Question	Marking Guidance	Mark	Comments
01.1	1. (There are) stretch-mediated sodium ion channels (in the membrane); 2. (Increased pressure) deforms/changes (sensory neurone/axon) membrane/lamella(e) OR (Increased pressure) deforms/changes sodium ion channels; 3. Sodium ion channels open; 4. Sodium ions diffuse in; 5. Depolarisation (leading to generator potential);	3 max (3 x AO1)	1. 2. 3. or 4. Accept Na ⁺ Max 2 if only reference to sodium and not sodium ions 4. Accept sodium ions move in down a concentration gradient 5. Accept inside becomes positive

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01.2	1. An action potential is only generated/produced when threshold is reached OR An action potential is not generated/produced until/unless threshold is reached; 2. When high pressure was applied (to the fingertip), threshold was reached OR High pressure is (an) above threshold (stimulus) OR High pressure generates an action potential;	2 (1 x AO1, 1 x AO3)	1. Ignore any numbers linked to threshold 1. and 2 Accept impulse for action potential 2. Accept references to 'the first peak' for high pressure 2. Accept the converse for low pressure

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01.3	(The) <u>refractory</u> (period);	1 (AO1)	Ignore any reference to absolute or relative Reject refraction period
